VANISHING OF H^0 OF THE PRODUCT TOTALIZATION OF A COMPLEX OF COMPLEXES WHOSE P-TH TERM HAS VANISHING H^{-P}

THE STACKS PROJECT AUTHORS

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Editorial note. The section lead-in at source lines 30092-30093, pointing to the discussion of double complexes in the Homology chapter, lies directly before the counted statement and is outside the counted spans; the card therefore needs no navigation stub, and the proof refers to no labelled item of any chapter.

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Allowed standard prerequisites (editorial).

- The differential on the product totalization Tot_pi of a double complex with entries A^{p,q} is, on elements of total degree n, the product over p + q = n of the components f^q_p + (-1)^p d^q_{A_p}. recalled at proof lines 30106-30111. Gives the explicit form of the single equation d(x) = 0 for a cocycle representative x = (x_p) in the product, which is then unwound degree by degree.
- A class in H^0 is represented by a cocycle of total degree zero, i.e. by a family x = (x_p) in the product of the A^{p,-p} with d(x) = 0. used at proof lines 30112-30115. Fixes the shape of the object that has to be exhibited as a boundary, so that the goal becomes finding y with d(y) = x.
- Exactness of a complex at a degree: if H^q(C) = 0 and an element z has d(z) = 0, then z is the boundary of some element of C^{q-1}. used at proof lines 30116-30121, first with H^0(A_0) = 0 and then with H^{-1}(A_1) = 0. Supplies the first element y_{-1} with d(y_{-1}) = x_0 and then the element y_{-2} that kills the corrected component x_1 + f_0(y_{-1}); the same step is iterated in the induction.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A_0^\bullet \to A_1^\bullet \to A_2^\bullet \to \ldots$ be a complex of complexes of abelian groups. Assume $H^{-p}(A_p^\bullet) = 0$ for all $p \geq 0$. Set $A^{p,q} = A_p^q$ and view $A^{\bullet,\bullet}$ as a double complex. Then $H^0(Tot_\pi(A^{\bullet,\bullet})) = 0$.*

Proof. Denote $f_p : A_p^\bullet \to A_{p+1}^\bullet$ the given maps of complexes. Recall that the differential on $\mathrm{Tot}_\pi(A^{\bullet,\bullet})$ is given by

$$\prod\nolimits_{p+q=n} (f_p^q + (-1)^p \mathrm{d}_{A_p^\bullet}^q)$$

on elements in degree n. Let $\xi \in H^0(\mathrm{Tot}_\pi(A^{\bullet,\bullet}))$ be a cohomology class. We will show ξ is zero. Represent ξ as the class of an cocycle $x = (x_p) \in \prod A^{p,-p}$. Since $\mathrm{d}(x) = 0$ we find that $\mathrm{d}_{A_0^\bullet}(x_0) = 0$. Since $H^0(A_0^\bullet) = 0$ there exists a $y_{-1} \in A^{0,-1}$ with $\mathrm{d}_{A_0^\bullet}(y_{-1}) = x_0$. Then we see that $\mathrm{d}_{A_1^\bullet}(x_1 + f_0(y_{-1})) = 0$. Since $H^{-1}(A_1^\bullet) = 0$ we can find a $y_{-2} \in A^{1,-2}$ such that $-\mathrm{d}_{A_1^\bullet}(y_{-2}) = x_1 + f_0(y_{-1})$. By induction we can find $y_{-p-1} \in A^{p,-p-1}$ such that

$$(-1)^p \mathrm{d}_{A_p^\bullet}(y_{-p-1}) = x_p + f_{p-1}(y_{-p})$$

This implies that $\mathrm{d}(y) = x$ where $y = (y_{-p-1})$. □